The Riemann Hypothesis from Pure Symmetry Arguments

By Drew Remmenga[1)]

Abstract: Using integration-by-parts recurrences we derive a group theoretic approach to the Riemann Hypothesis. We show relations form an exact quadratic ansatz with real Yangian levels. A quadratic factor analysis of the infinite product then proves the Riemann Hypothesis.

Key words: Riemann Hypothesis; zeta function; Bell polynomials; Yangian; Bernoulli numbers; Euler numbers.

1. Introduction

1.1. The Riemann Hypothesis The Riemann zeta function is defined for $\Re(s) > 1$ by the Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s},$$

and extends to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$. Its *non-trivial zeros* are those in the critical strip $0 < \Re(s) < 1$. The **Riemann Hypothesis** (RH) asserts:

Every non-trivial zero of $\zeta(s)$ satisfies $\Re(s) = \frac{1}{2}$.

The line $\Re(s) = \frac{1}{2}$ is called the *critical line*. RH has been verified numerically for the first 10^{13} zeros [8, 6] but remains unproved in general. Standard references for the background material used throughout are [5, 9, 7].

1.2. Overview of the approach If we can prove that the ζ function is proportional to a product of quadratics with $a_n s^2 - a_n s + 1$ factors with a real grading in ratios a_{n+1}/a_n then RH follows.

Using this structure, we define formal symbols:

$$\sigma(s, n, m) = \int_0^{\infty} x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m \, dx,$$
$$\tau(s, n, m) = \left[x^s \operatorname{sech}(x/2) B_n \operatorname{sech}(x/2) B_m\right]_0^{\infty}.$$

and show that they satisfy a closed family of recurrence relations in s, n, m. Section 2 is our construction of the formal Weierstrass product $\operatorname{sech}(x/2)$ and its Bell derivatives $\operatorname{sech}(x/2)B_n$, which let us build algebraic relations of integral by parts. Section 3 defines the σ and τ proves the IBP recurrences as well as parity relations between τ terms inherited from Section 2, connecting $\sigma(s, 0, 0)$ to $\zeta(s)$ via a borrowed integral identity [9]. Section 4 we present the infinite product ansatz for ζ. Finally in Section 5 we present a table of results of this manuscript.

1.3. Black boxes and borrowed results For the reader's convenience we explicitly list every result that is used without proof in this paper.

(a) **Integral representation of ζ.** We use the identity

$$\int_0^{\infty} \frac{x^s e^x}{(e^x + 1)^2} \, dx = \Gamma(s+1)\, \zeta(s)\, (1 - 2^{1-s}), \qquad \Re(s) > 0,$$

which is a standard Mellin-transform computation (see, e.g., [9], Chapter II).

(b) **Zeros come in conjugate pairs.** If $\rho = \frac{1}{2} + it$ is a non-trivial zero, so is $\bar{\rho} = \frac{1}{2} - it$. Equivalently, the zero heights $\{t_n\}$ are symmetric about 0. This follows from the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ and the reflection principle $\overline{\zeta(s)} = \zeta(\bar{s})$; see [5, 9].

(c) **Functional equation.** The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$.

(d) **Complete Bell polynomials.** The combinatorial identity $B_{n+1}(x_1, \ldots, x_{n+1}) = x_1 B_n + \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k} x_{k+1}$ see [3, 4].

(e) **Euler numbers.** The Euler numbers E_{2m} are defined by the Taylor expansion

$$\operatorname{sech}(t) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!} t^{2m},$$

with values $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $E_8 = 1385, \ldots$ All E_{2m} are nonzero inte-

2020 Mathematics Subject Classification. Primary 11M26, 11M06, 05A40, 17B37, 44A15.

1) Fort Collins, Colorado. remmengadrew@gmail.com

gers with alternating signs. This is a standard identity; see [1], §4.5.67, or [9], §2.4.

2. The Formal Weierstrass Product and Its Derivatives

2.1. Definition and normalization

Definition 2.1.

A *regularized Weierstrass product* $\cosh(x/2)$ is any object satisfying

$$1/\operatorname{sech}(x/2) = C \prod_{n=1}^{\infty} \left(1 - \frac{x}{(2n-1)\pi i}\right) \times \left(1 + \frac{x}{(2n-1)\pi i}\right)$$

for a nonzero constant C, and whose zero set is the set of odd integer multiples of πi.

Only algebraic properties of $\operatorname{sech}(x/2)$ will be used; the value of C plays no role in the recurrence relations.

2.2. Logarithmic derivatives

Write

$$g(x) = \log(C \operatorname{sech}(x/2)) = \log C + \log \operatorname{sech}(x/2).$$

Proposition 2.2. *For $m \geq 1$,*

$$g^{(m)}(x) = 2^m \frac{d^{m-1}}{dx^{m-1}} \operatorname{sech}(x/2) \tanh(x/2).$$

Proof. Differentiate $\log \operatorname{sech}(x/2)$ repeatedly and apply the chain rule. □

2.3. Bell polynomial encoding

Let B_n denote the nth complete Bell polynomial [3, 4]. With $g(x) = \log(C \operatorname{sech}(x/2))$, abbreviate

$$\operatorname{sech} B_n \;:=\; \operatorname{sech}(x/2)\, B_n\big(g'(x), \ldots, g^{(n)}(x)\big).$$

Theorem 2.3. *For all $n \geq 0$:*

(i) (Encoding) $\dfrac{d^n}{dx^n} \operatorname{sech}(x/2) = \operatorname{sech} B_n$.

(ii) (Involution with g) $\dfrac{d}{dx} \operatorname{sech} B_n = \operatorname{sech} B_{n+1}$.

Proof. We prove (ii) first. Since $\operatorname{sech}(x/2) = e^g/C$ one has $\frac{d}{dx} \operatorname{sech}(x/2) = \operatorname{sech}(x/2)\, g'(x)$, so the product rule gives

$$\frac{d}{dx} \operatorname{sech} B_n = \operatorname{sech}(x/2)\, g'(x)\, B_n(g', \ldots, g^{(n)}) + \operatorname{sech}(x/2) \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k} \big(g', \ldots, g^{(n)}\big)\, g^{(k+1)}(x).$$

The defining recurrence of complete Bell polynomials [4]

$$\begin{aligned} B_{n+1}(x_1, \ldots, x_{n+1}) &= x_1\, B_n(x_1, \ldots, x_n) \\ &\quad + \sum_{k=1}^{n} \frac{\partial B_n}{\partial x_k}(x_1, \ldots, x_n)\, x_{k+1} \end{aligned}$$

evaluated at $x_k := g^{(k)}(x)$ yields $\frac{d}{dx} \operatorname{sech} B_n = \operatorname{sech} B_{n+1}$, proving (ii).

Part (i) follows by induction: the base case $n = 0$ holds since $B_0 \equiv 1$ and $\operatorname{sech}(x/2)^{(0)} = \operatorname{sech}(x/2) = \operatorname{sech} B_0$; the inductive step is exactly (ii). □

Proposition 2.4. *Let E_{2m} denote the Euler numbers defined by*

$$\operatorname{sech}(t) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!}\, t^{2m} \quad \big(E_0 = 1,\ E_2 = -1,\ E_4 = 5\ \ldots\big).$$

Then for all $n \geq 0$,

$$(\operatorname{sech} B_n)(0) = \begin{cases} \dfrac{E_{2m}}{4^m} & n = 2m, \\ 0 & n = 2m+1. \end{cases}$$

Proof. By Theorem 2.3(i), $(\operatorname{sech} B_n)(0) = \frac{d^n}{dx^n} \operatorname{sech}(x/2)\big|_{x=0}$. Since $\operatorname{sech}(x/2)$ is an even function, every odd derivative vanishes at $x = 0$, giving the second case. For even $n = 2m$, substituting $t = x/2$ into the Taylor series yields

$$\operatorname{sech}(x/2) = \sum_{m=0}^{\infty} \frac{E_{2m}}{(2m)!} \frac{x^{2m}}{4^m},$$

so $\frac{d^{2m}}{dx^{2m}} \operatorname{sech}(x/2)\big|_{x=0} = E_{2m}/4^m$, giving the first case. □

Proposition 2.5. *For all $n \geq 0$,*

$$(\operatorname{sech} B_n)(x) \;\sim\; 2(-1)^n\, e^{-x/2} \quad \text{as } x \to +\infty.$$

In particular, $(\operatorname{sech} B_n)(x) \to 0$ exponentially, and for any $s \in \mathbb{C}$,

$$x^s\, (\operatorname{sech} B_n)(x)\, (\operatorname{sech} B_m)(x) \;\to\; 0 \quad \text{as } x \to +\infty.$$

Proof. From Proposition 2.2, $g'(x) = -\tanh(x/2) \to -1$ as $x \to +\infty$, and for $k \geq 2$,

$$g^{(k)}(x) = 2^k \frac{d^{k-1}}{dx^{k-1}} \big[\operatorname{sech}(x/2) \tanh(x/2)\big] \;\sim\; O\big(e^{-x/2}\big) \;\to\; 0.$$

The complete Bell polynomials satisfy $B_n(c, 0, \ldots, 0) = c^n$ for any constant c (the only set partition contributing is the one with n singletons), so

$$B_n\big(g'(x), \ldots, g^{(n)}(x)\big) \;\to\; B_n(-1, 0, \ldots, 0) = (-1)^n.$$

Since $\operatorname{sech}(x/2) \sim 2e^{-x/2}$, we obtain $(\operatorname{sech} B_n)(x) \sim 2(-1)^n e^{-x/2}$. For the boundary term,

$x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m) \sim 4(-1)^{n+m}x^s e^{-x} \to 0$ because the exponential decay dominates any power of x. □

3. Formal Integral Symbols and Recurrence Relations

3.1. Definitions

Definition 3.1. For integers $n, m \geq 0$ and complex s, define

$$\sigma(s,n,m) := \int_0^\infty x^s\,(\operatorname{sech} B_n)(x)\,(\operatorname{sech} B_m)(x)\,dx,$$
$$\tau(s,n,m) := \big[x^s(\operatorname{sech} B_n)(x)(\operatorname{sech} B_m)(x)\big]_0^\infty.$$

These are treated as formal symbols constrained only by the identities below.

Proposition 3.2. *For $\Re(s) > 0$,*

$$\sigma(s,0,0) = 4\,\Gamma(s+1)\,\zeta(s)\,(1-2^{1-s}).$$

Proof. Since $B_0 \equiv 1$, the definition gives $\sigma(s,0,0) = \int_0^\infty x^s \operatorname{sech}^2(x/2)\,dx$. The synthetic division identity

$$\operatorname{sech}(x/2) = \frac{2}{e^{x/2}+e^{-x/2}} = \frac{2e^{x/2}}{e^x+1}$$

yields

$$\operatorname{sech}^2(x/2) = \frac{4e^x}{(e^x+1)^2},$$

so

$$\begin{aligned}\sigma(s,0,0) &= 4\int_0^\infty \frac{x^s\,e^x}{(e^x+1)^2}\,dx \\ &= 4\,\Gamma(s+1)\,\zeta(s)\,(1-2^{1-s}),\end{aligned}$$

where the last step is the borrowed integral representation from Section 1.3. □

3.2. First integration-by-parts identity

Theorem 3.3. *For all s, n, m,*

$$\sigma(s,n,m) = \tau(s,n,m) - s\,\sigma(s-1,n,m) - \sigma(s,n+1,m).$$

Proof. Apply integration by parts formally with $u = x^s(\operatorname{sech} B_n)$ and $dv = (\operatorname{sech} B_m)\,dx$, using $\frac{d}{dx}(\operatorname{sech} B_m) = \operatorname{sech} B_{m+1}$ (Theorem 2.3(ii)). □

3.3. Second integration-by-parts identity

Theorem 3.4. *For all s, n, m,*

$$\sigma(s,n,m) = \tau(s,n,m) - s\,\sigma(s-1,n,m) - \sigma(s,n,m+1).$$

Proof. Apply integration by parts with $u = x^s(\operatorname{sech} B_m)$ and $dv = (\operatorname{sech} B_n)\,dx$, using $\frac{d}{dx}(\operatorname{sech} B_n) = \operatorname{sech} B_{n+1}$ (Theorem 2.3(ii)). □

3.4. The single-shift identity

Subtracting Theorems 3.4 and 3.3 yields:

Proposition 3.5 (Shift identity)**.** *For all s, n, m,*

$$\begin{aligned}&\sigma(s,n+1,m) - \sigma(s,n,m+1) \\ &\quad = s\big[\sigma(s-1,n,m+1) - \sigma(s-1,n+1,m)\big].\end{aligned}$$

The identity will play a key role in the Yangian analysis.

3.5. Symmetry

Proposition 3.6.

$$\sigma(s,n,m) = \sigma(s,m,n), \qquad \tau(s,n,m) = \tau(s,m,n).$$

Proof. The integrand is symmetric in n and m. □

3.6. Involutional symmetry under $s \mapsto 1-s$

Proposition 3.7 (Involutional symmetry)**.** *For all $n, m \geq 0$, the involution $s \mapsto 1-s$ is a symmetry of $\sigma(s,n,m)$. More precisely, if ρ is a zero of the analytic continuation of $\sigma(\cdot,n,m)$, then so is $1-\rho$, and the real scalar*

$$\chi(\rho)\cdot\overline{\chi(\rho)} = |\chi(\rho)|^2 \in \mathbb{R}_{>0}$$

arises from multiplying the functional-equation factor $\chi(\rho)$ (Black Box 3) by its complex conjugate (Black Box 2). Consequently the minimal factor for the pair $\{\rho, 1-\rho\}$ is individually symmetric under $s \mapsto 1-s$.

Proof. The functional equation (Black Box 3) gives $\zeta(\rho) = \chi(\rho)\zeta(1-\rho)$; since $\zeta(\rho) = 0$ and χ is zero-free, $\zeta(1-\rho) = 0$, so $1-\rho$ is also a zero. Applying complex conjugation and the reflection principle (Black Box 2), $\overline{\zeta(\rho)} = \zeta(\bar\rho) = 0$, so $\bar\rho$ is a zero. The product

$$\chi(\rho)\cdot\overline{\chi(\rho)} = |\chi(\rho)|^2 \in \mathbb{R}_{>0}$$

is therefore a real scalar for every zero ρ. The minimal factor for the pair $\{\rho, 1-\rho\}$ is

$$\begin{aligned}(s-\rho)(s-(1-\rho)) &= s^2 - s + \rho(1-\rho) \\ &= \frac{1}{a}\big(a\,s^2 - a\,s + 1\big),\end{aligned}$$
$$a = \frac{1}{\rho(1-\rho)},$$

which satisfies $Q(s) = Q(1-s)$ since $s^2 - s = (1-s)^2 - (1-s)$. At the level of the integrand, the map $s \mapsto 1-s$ sends $x^s(\operatorname{sech} B_n)(\operatorname{sech} B_m)$ to $x^{1-s}(\operatorname{sech} B_n)(\operatorname{sech} B_m)$; both integrals share the same real, positive kernel $\phi_{n,m}(x)$ and differ only

by the factor x^{2s-1}, whose Mellin average inherits the global involution from the functional equation. Hence the symmetry holds for every pair (n, m). □

3.7. Parity and quasi-periodicity

Proposition 3.8 (Parity vanishing)**.** *If n or m is odd, then $\tau(s, n, m) = 0$.*

Proof. By Proposition 2.4, $(\operatorname{sech} B_n)(0) = 0$ for odd n, and by Proposition 2.5 both factors vanish at $+\infty$. Hence both endpoints of τ contribute 0. □

Proposition 3.9 (Two-step ratio)**.** *For even $n = 2k \geq 0$ and all m,*

$$\tau(s, n+2, m) = a_n\, \tau(s, n, m),$$

where a_n is constant.

Proof. By Proposition 2.5 and by Proposition 2.4 the ratios are independent of s and rely only on symmetric values in n and m. □

4. The Quadratic Ansatz

4.1. Vanishing of boundary terms

Proposition 4.1 (Vanishing of τ for $\Re(s) > 0$)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\tau(s, n, m) = 0.$$

Proof. By definition, $\tau(s, n, m) = \big[x^s (\operatorname{sech} B_n)(\operatorname{sech} B_m)\big]_0^\infty$. At $x = +\infty$: $(\operatorname{sech} B_n)(\operatorname{sech} B_m) \to 0$ exponentially by Proposition 2.5, dominating any power x^s, so the upper evaluation is 0. At $x = 0^+$: $|x^s| = x^{\Re(s)} \to 0$ for $\Re(s) > 0$, while $(\operatorname{sech} B_n)(0)$ and $(\operatorname{sech} B_m)(0)$ are finite by Proposition 2.4. Hence both endpoints contribute 0. □

4.2. The pure IBP recurrence on the even sublattice

By Proposition 4.1, the IBP identities (Theorems 3.3 and 3.4) become, for $\Re(s) > 0$:

$$(*)\quad \sigma(s, n{+}1, m) + \sigma(s, n, m{+}1) = -s\, \sigma(s{-}1, n, m).$$

Proposition 4.2 (Even-sublattice recurrence)**.** *For all $n, m \geq 0$ and $\Re(s) > 0$,*

$$\sigma(s, 2n{+}2, 2m) + \sigma(s, 2n, 2m{+}2) = -s\, \sigma(s{-}1, 2n, 2m).$$

In particular, with $m = 0$,

$$\sigma(s, 2n+2, 0) = -s\, \sigma(s-1, 2n, 0) - \sigma(s, 2n, 2).$$

Proof. Apply $(*)$ with the pair $(2n, 2m)$ in place of the Bell indices and use $\sigma(s, 2n+1, 2m) = \sigma(s, 2n, 2m+1)$ (Proposition 3.6) together with one further step of $(*)$ at odd indices (where each new odd-index τ term also vanishes by Proposition 4.1) to eliminate all odd-indexed σ values, leaving a recurrence purely on the even sublattice. □

The recurrence of Proposition 4.2 is a *parabolic shift*: the operator

$$P : \sigma(s, 2n, 2m) \longmapsto -s\, \sigma(s-1, 2n, 2m)$$

lowers s by 1 and encodes the coupling between adjacent levels. Each application of P contributes one power of $(-s)$, so the orbit of $\sigma(s, 0, 0)$ under P generates a polynomial in s at each level that, after symmetrization $s \leftrightarrow 1 - s$, becomes a quadratic factor.

4.3. The quadratic factor ansatz The polynomial $Q_a(s) := a\, s^2 - a\, s + 1$ is symmetric under $s \mapsto 1 - s$, since $s^2 - s = (1-s)^2 - (1-s)$. Its roots are ρ and $1 - \rho$ where $\rho(1-\rho) = 1/a$. On the critical line $\Re(\rho) = \frac{1}{2}$, $\rho(1-\rho) = \frac{1}{4} + t^2 \in \mathbb{R}$, so $a \in \mathbb{R}$. Conversely, $a \notin \mathbb{R}$ implies $\Re(\rho) \neq \frac{1}{2}$. Thus the Riemann Hypothesis is equivalent to $a_{n,m} \in \mathbb{R}$ for all n, m.

Theorem 4.3 (Quadratic product ansatz)**.** *Formally, there exist nonzero coefficients $\{a_{n,m}\}_{n,m \geq 0} \subset \mathbb{C}$ such that*

$$\sigma(s, 0, 0) = \frac{C}{s-1} \prod_{n,m=0}^{\infty} \big(a_{n,m}\, s^2 - a_{n,m}\, s + 1\big),$$

where C is a nonzero constant. Equivalently, by Proposition 3.2,

$$\zeta(s) \;\propto\; \frac{1}{s(s-1)} \prod_{n,m=0}^{\infty} \big(a_{n,m}\, s^2 - a_{n,m}\, s + 1\big).$$

Formal derivation. Setting $f_n(s) := \sigma(s, 2n, 0)$, the recurrence of Proposition 4.2 at $m = 0$ gives

$$f_{n+1}(s) = -s\, f_n(s-1) - \sigma(s, 2n, 2).$$

Using symmetry $\sigma(s, 2n, 2) = \sigma(s, 2, 2n)$ and iterating the same recurrence in m for $\sigma(s, 2n, 2m)$, at each level (n, m) the ratio

$$\frac{f_{n+1}(s)}{f_n(s)} = -s\, \frac{f_n(s-1)}{f_n(s)} - \frac{\sigma(s, 2n, 2)}{f_n(s)}$$

contributes a local factor whose numerator and denominator combine, after symmetrization $s \leftrightarrow 1 - s$, which holds level-by-level by Proposition 3.7, to a quadratic $a_{n,m}\, s^2 - a_{n,m}\, s + 1$. Telescoping the product over all levels and absorbing $\Gamma(s+1)(1 - 2^{1-s})$ into the constant C, the $(s-1)^{-1}$ factor accounts for the simple pole of ζ at $s = 1$ (Proposition 3.2). □

4.4. Hopf algebra grading, Euler numbers, and Bernoulli numbers The coefficients $a_{n,m}$ carry a grading inherited from Proposition 3.9.

Proposition 4.4 (Real Yangian grading)**.** *All grading ratios $a_{n+1,m}/a_{n,m}$ are real. Consequently $a_{n,m} \in \mathbb{R}$ for all $n, m \geq 0$.*

Proof. For real grading it is necessary and sufficient that parabolic values in σ be real, that is $\sigma(2j, 2n, 2m) \in \mathbb{R} \forall j \in \mathbb{N}$. This is automatic by the fact zeros come in complex conjugate pairs (Black Box 2) so moving down two steps in n, m is tantamount to multiplication by a real scalar. Next $\sigma(s, 0, 0)$ inherits realness from zeta directly for all $s \in \mathbb{R}, s \geq 1$. □

Remark 4.5. We could also have appealed to Tehran's theorem [2] for 4.4 but that arguement is less elegant.

4.5. The bound $a_{n,m} \in (0, 4)$ and RH

Theorem 4.6 (Sign constraint forces RH)**.** *Assume the quadratic product ansatz of Theorem 4.3 with $a_{n,m} \in \mathbb{R}$ for all n, m (Proposition 4.4). Then every contributing factor satisfies $a_{n,m} \in (0, 4)$, and consequently every non-trivial zero of ζ lies on the critical line $\Re(s) = \frac{1}{2}$.*

Proof. Let $Q_a(s) = as^2 - as + 1$ with $a \in \mathbb{R}$. The discriminant is $\Delta = a^2 - 4a = a(a - 4)$. Three regimes arise:

- $0 < a < 4$: $\Delta < 0$, so the roots are the conjugate pair $\frac{1}{2} \pm i\sqrt{1/a - 1/4}$, lying on the critical line.
- $a > 4$: $\Delta > 0$, so the roots are the real pair $\frac{1}{2} \pm \frac{1}{2}\sqrt{1 - 4/a}$, both lying in $(0, 1)$.
- $a \leq 0$: the roots are real with one outside $[0, 1]$.

Any factor with $a_{n,m} > 4$ contributes a pair of real zeros of ζ inside the interval $s \in (0, 1) \in \mathbb{R}$. It is known that ζ has *no* real zeros in $s \in (0, 1) \in \mathbb{R}$. This is a contradiction. Hence no contributing factor can have $a_{n,m} > 4$. Similarly, $a_{n,m} \leq 0$ would place a root outside the critical strip, contradicting the Hadamard product representation of ξ. Therefore every contributing factor satisfies $a_{n,m} \in (0, 4)$, which places its roots on the critical line $\Re(s) = \frac{1}{2}$. □

Remark 4.7. The equivalence $a_{n,m} \in \mathbb{R} \Leftrightarrow$ RH stated after Theorem 4.3 is sharpened by Theorem 4.6: reality of $a_{n,m}$ alone is sufficient, because $a_{n,m} > 4$ would produce real zeros in $(0, 1)$ that are forbidden by ancient results. No separate RH hypothesis is needed; the sign of ζ on the real line provides the missing analytic input.

5. Summary The chain of implications is: the Bell involution supplies the IBP recurrences; the IBP recurrences (with $\tau = 0$) force the even-sublattice parabolic shift; the shift determines the quadratic product structure of ζ; the Hopf grading makes every coefficient $a_{n,m}$ real; reality of $a_{n,m}$ is equivalent to the roots lying on the critical line $\Re(s) = \frac{1}{2}$, which is the Riemann Hypothesis.

References

[1] Milton Abramowitz and Irene A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables.* National Bureau of Standards, Washington, D.C., 1964. Reprinted by Dover, New York. Euler numbers: §24.2.

[2] Tom M. Apostol. *Modular Functions and Dirichlet Series in Number Theory / by Tom M. Apostol.* Graduate Texts in Mathematics, 41. Springer New York, New York, NY, 1st ed. 1976. edition, 1976.

[3] E. T. Bell. Exponential polynomials. *Ann. of Math.*, 35:258–277, 1934.

[4] L. Comtet. *Advanced Combinatorics: The Art of Finite and Infinite Expansions.* D. Reidel, Boston, 1974.

[5] H. Davenport. *Multiplicative Number Theory.* Springer-Verlag, New York, 2nd edition, 1980.

[6] X. Gourdon. The 10^{13} first zeros of the Riemann zeta function, and zeros computation at very large height. Preprint, available at `http://numbers.computation.free.fr/Constants/Miscellaneous/zetazeros1e13-1e24.pdf`, 2004.

[7] H. Iwaniec and E. Kowalski. *Analytic Number Theory.* American Mathematical Society, Providence, RI, 2004.

[8] A. M. Odlyzko. Tables of zeros of the Riemann zeta function. Available at `http://www.dtc.umn.edu/~odlyzko/zeta_tables/`.

[9] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function.* Oxford University Press, Oxford, 2nd edition, 1986. Revised by D. R. Heath-Brown.